SyncQuest

How Two People on Different Devices Use It

This guide covers the one scenario that needs a small extra setup step: two separate physical devices (not two tabs on the same laptop) joining the same realm over Wi-Fi. `localhost` only works within a single machine, so both devices need to reach the host machine by its actual network address instead.

1. **Run both servers on one machine (the "host")**

   Pick one of the two devices (or a third machine) to run the app's servers. From the `p2p-editor/` project folder, in two separate terminals:

   ```
   # Terminal 1
   npm run signaling

   # Terminal 2
   npm run dev
   ```

2. **Find the host machine's LAN IP address**

   On Windows, open a terminal on the host and run:

   ```
   ipconfig
   ```

   Look for **IPv4 Address** under the active Wi-Fi adapter -- e.g. `192.168.1.23`. Both devices must be connected to the same Wi-Fi network as this host.

3. **Point the frontend at that IP instead of localhost**

   Edit `frontend/.env` on the host (copy from `.env.example` if it doesn't exist yet):

   ```
   VITE_SIGNALING_URL=ws://192.168.1.23:4444
   ```

   Use the host's actual IP from Step 2. Then restart the dev server (`npm run dev`) so it picks up the change. The dev server is already configured with `host: true` in `vite.config.js`, so it accepts connections from other devices on the network, not just the host itself.

4. **Open the app on device 1 and copy the Portal Link**

   On device 1 (can be the host machine itself), open:

   ```
   http://192.168.1.23:5173
   ```

   Note this is the LAN IP, **not** `localhost`. A realm code is generated automatically. Click **Copy Portal Link** to grab the shareable URL.

5 **Open the same link on device 2**

Send that link to device 2 (a phone, another laptop, any browser -- same Wi-Fi network) and open it there. Both devices need to be able to reach the host on ports `5173` (frontend) and `4444` (signaling).

6 **Allow the firewall prompt on first run**

If Windows asks whether to allow Node.js through the firewall the first time you start either server, choose **Allow** for Private/Home networks. If you deny it, device 2's requests get silently blocked before they ever reach the app.

What you should see when it works

- Both devices show the **"Portal Open -- Synced!"** status badge.
- The **Party panel** on each device lists the other device's avatar/name.
- Typing on either device appears on the other within about a second.
- Each device's cursor is visible, highlighted in its own color, inside the other's editor.

If it doesn't work

> **Note:** If both devices are confirmed on the same Wi-Fi but the status stays stuck on "Connecting...", the most common cause is **router client isolation** (common on guest/public Wi-Fi networks) -- it deliberately blocks device-to-device traffic, which is a network setting outside the app's control, not a bug. Try a home/private network, or a personal mobile hotspot, instead.

- Double-check `VITE_SIGNALING_URL` uses the host's current LAN IP -- this can change if the host reconnects to Wi-Fi.
- Confirm the signaling server terminal is still running and shows no errors.
- Confirm both devices used the LAN-IP URL, not `localhost`, to open the app in the first place.

See `README.md` in the project folder for the full setup guide, deployment steps, and the complete testing checklist.